

ZALAK RAJVANSHI

AI / ML Engineer

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OVERVIEW

AI/ML Engineer with hands-on experience shipping production systems across **generative AI**, computer vision, NLP and LLM-driven automation. I build end-to-end (from model training and embedding pipelines to deployed web products) and focus on turning complex AI capabilities into measurable, real-world impact.

EDUCATION

Silver Oak University (SOCET)

B.Tech, Computer Engineering · CGPA: 8.83 / 10

2022 – 2026

Ahmedabad, India

EXPERIENCE

The Product Folks

Feb 2026 – Jun 2026

AI Engineering Intern

Ahmedabad, India

AI Content Engine | Python, Higgsfield MCP, LLM Orchestration, Media Pipelines

- Designed and shipped an AI content production pipeline via **Higgsfield MCP** that converts plain-text briefs into ready-to-publish videos and images. Generated **200+ branded creatives** for community campaigns and marketing at a fraction of the time and cost of a manual creative pipeline.
- Owned the LLM-driven prompt orchestration, automated post-processing and format-aware asset routing across short-form video and image variants. Cut per-asset turnaround from **days to minutes (~95% reduction)**.

Semantic Talent Discovery Engine | Python, Embeddings, Semantic Search, Next.js | tpf-profile-matcher.vercel.app

- Built an AI candidate-matching platform that ranks best-fit profiles from a **23,000+ indexed professional pool** against any uploaded job description. Turns what used to be multi-day manual screening into a seconds-long semantic search.
- Designed the embedding pipeline, JD parsing, similarity scoring and filter-aware ranking; deployed as a production Next.js app delivering **top-ranked candidate lists in <1 second per query** and cutting screening time from **days to minutes per role**.

AI-Driven Lead Generation & Outreach Engine | Python, Selenium, LangChain, LLM

- Developed an end-to-end AI outreach engine combining two modules: the **Lead Generation Module** auto-discovers LinkedIn profiles, parses their career history and ranks them against the active job description; the **Lead Outreach Module** then runs a structured **3-message WhatsApp sequence with 36-hour follow-up gaps**, cutting manual sourcing-to-engagement effort by **~70%**.
- Added **LLM-powered reply classification** (~85% 3-class accuracy across positive / negative / neutral), a daily summary bot for HR visibility and persisted-state recovery; designed to scale to **100+ candidates per campaign** end-to-end without human handoff.

Arova · Conversational AI Platform | Python, JavaScript, LangChain | arova-tpf.vercel.app

- Delivered Arova, a production conversational AI platform powering **24/7 community engagement** across **8-10 high-frequency intent flows**. Cuts response latency from **hours to under 2 seconds**, eliminates **~80% of manual triage effort** and orchestrates end-to-end user conversations without human handoff.
- Composed the system across three layers: a JavaScript conversational UI, a Python **LangChain**-powered backend for intent routing and conversation-memory handling, plus a separate AI processing pipeline for asynchronous inference. Implemented prompt engineering and graceful fallback for low-confidence or out-of-scope queries.

PROJECTS

Sign Language Detection (ASL → Speech) | Python, TensorFlow, MediaPipe, OpenCV, pyttsx3 | GitHub ↗

- Real-time ASL recognition covering **29 classes**; **Conv1D CNN** trained on **30K+ MediaPipe-extracted feature frames** reaches **98.78% validation accuracy** at **~25-30 FPS** (sub-40 ms/frame) on CPU, with stability filtering (7-10 frames, >88%), TextBlob spell correction and pyttsx3 TTS for full-sentence speech output.

Gitlane · AI Commits, Secret Guard & GitHub Coach | *Python, TypeScript, Flask, SQLite, LLM, VS Code Extension API, GitHub API* | [GitHub ↗](#)

- AI dev tool spanning **CLI, browser dashboard and VS Code extension**; writes **Conventional-Commits** messages from `git diff`, auto-scans staged files for leaked API keys and remediates them by moving secrets to `.env` and rewriting source with `os.getenv / process.env`, plus tracks daily shipping streaks, weekly sprints and per-repo goals via a shared SQLite memory. Cuts manual commit effort by **~80%** and prevents pre-push secret leaks; open-source (MIT) with cross-platform installer.

SmartTask AI · Context-Aware Day Planner | *React, Vite, Tailwind, Node, Express, PyTorch, Hugging Face Transformers, JWT* | [GitHub ↗](#)

- Full-stack productivity app converting natural-language goals into time-blocked schedules in **<3 s**; cuts manual day-planning effort by **~60%**, JWT auth + externalised model weights for AWS / Render deploys.

Smart Notes AI · Handwriting to Structured Notes | *Python, OCR, Layout Analysis, JavaScript* | [GitHub ↗](#)

- Pipeline converting handwritten pages into searchable, structured digital notes via OCR + layout analysis + semantic classification; **~85% line-level extraction accuracy**, cuts manual digitisation effort by **~75%**.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, SQL, HTML / CSS
Frameworks & Libraries: TensorFlow, PyTorch, Keras, Hugging Face Transformers, scikit-learn, MediaPipe, OpenCV, LangChain, React, Next.js, FastAPI, Flask, Node.js, Express.js, Tailwind CSS
AI / ML & Data: Generative AI, Computer Vision, NLP, OCR, LLM Agents, MCP, RAG, Embeddings, Semantic Search, Prompt Engineering, Model Deployment, Pandas, NumPy, SQL, MongoDB, SQLite
Tools & Cloud: Docker, Git & GitHub, Hugging Face Hub, Selenium WebDriver, n8n, VS Code Extension API, AWS (foundational), JWT, REST APIs

CERTIFICATIONS

Foundations of AI & ML, Microsoft · Nov 2024 | **Gen AI Study Jams 2024**, Google Cloud Skills Boost · Nov 2024

LEADERSHIP & COMMUNITY

Vice Chairperson, *IEEE SOU WIE SB AG* (2025–26): Led technical and professional initiatives across the chapter; coordinated multi-event execution and championed women-in-engineering programming on campus.
Senior Curation Executive, *IEEE SOU Student Branch* (2026): Curated technical events, speaker sessions and knowledge-sharing activities; owned content quality and run-of-show for chapter programming.
Content Lead, *GDG On Campus SOU* (2024–25): Owned the chapter's technical content strategy and produced developer-facing material that drove community engagement and event attendance.
Community Volunteer, *AWS Cloud Clubs SOU* (2024–Present): Co-organized cloud workshops and hands-on AWS labs, onboarding students into core cloud services.